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Fuse Plugin Guide and Reference

Blackmagicdesign

Fusion SDK



Leading the Creative Video Revolution

About this Document

This document is divided into two sections: The Fuse Guide and the Fuse Reference. The first section, the Fuse Guide, explains the fuse application programming interface (API). Fuses are like plugins that can be developed easily without the need of compilers and development environments. Fuses can go beyond Macros, and their source code can easily be converted into C++ source for the Fusion SDK. Resolve and Fusion has a built in compiler and will compile on the fly. Access to GPU processing can easily be done in a Fuse as well as utilizing optimized core processing functions already built into Fusion

This guide contains information on how to get started, how the API is laid out to represent the application model, and how to deal with it in practice. The first section refers to example fuses that ship with this guide The second section, the Fuse Reference, assumes you have an understanding of the programming concepts and the fundamentals of the first section. The Fuse Reference describes the common API, its objects, members, arguments and usage.

Target Audience

This document is intended for developers, technical directors, and users knowledgeable in programming. It was by no means written to teach programming concepts and does not act as a reference for programming languages. Please refer to Lua.org for Further language references.

If you are capable of editing Macros in a text editor and adding Scripting functions to customize tooling, then Fuses extend further with more functionality and lower level capabilities.

Requirements

In order to follow this guide, you will need to have a copy of Blackmagic Design Fusion or Resolve installed. The source code of both fuse languages needs to be stored as plain text, which can be written in any non-formatting text processor like Notepad or TextEdit. It is recommended to make use of a dedicated code editor to benefit from syntax highlighting and language-specific features, like SciTE.

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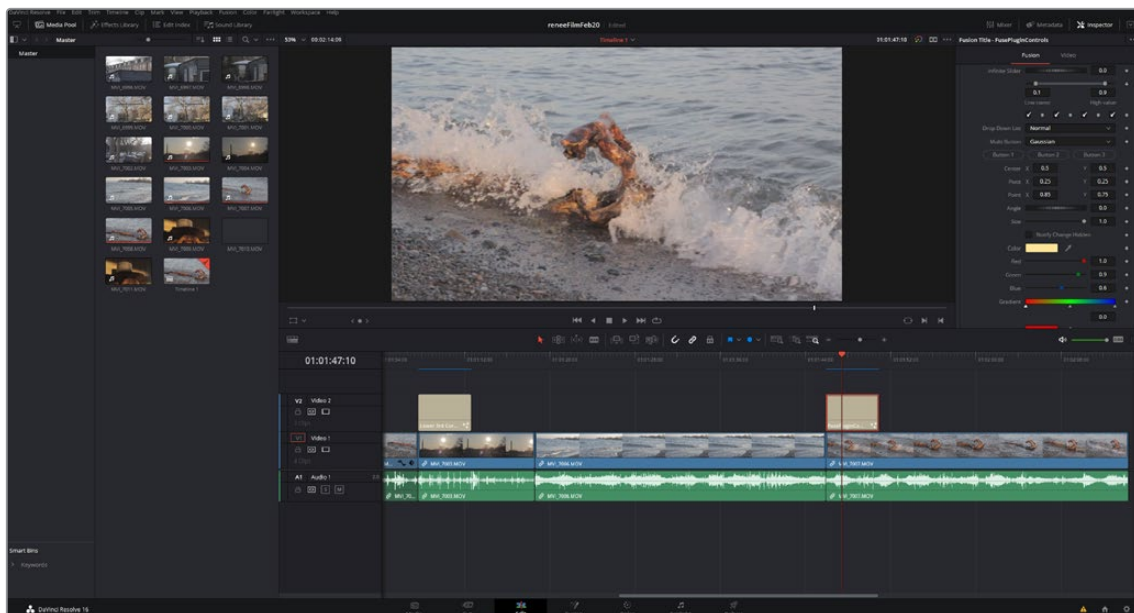
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About Fuse Plugins

- Fuses are like plugins that are hosted by Resolve-Fusion engine and Fusion Studio can be developed easily in a Text editor without the need of compilers and development environments.
- Builtin UI toolbox, with many different controllers, onscreen widgets are available to use.
- Development is rapid as Fuses can be developed and reloaded on the fly without restarting Resolve and Fusion. Test, Edit, and Reload the updated source code will compile on the fly and run.
- There is also a builtin core of image processing functions like Blur, Merge, Color operations and Image operation that utilize optimized core processing and GPU processes.
- Fuses can be multithreaded and GPU processing like DCTL can be used.
- Fuse source code uses Lua, which is a C like programming language, and their source code can easily be converted into C++ source for the Fusion SDK.
- The JustInTime (JIT) flavor of Lua that is utilized in Fusion which compiles on the fly for performance.



Installation

There are 7 Template Example Fuses installed in the Developer section of Resolve and Fusion with inline documentation (comments). This section will offer an overview of basic Fuse programming and refer to these examples for more in depth study.

First copy these fuses from the developer directory and paste into the these directories

Resolve

- **macOS:** ~/Library/Application Support/Blackmagic Design/DaVinci Resolve/Support/Fusion/Fuses
- **Windows:** C:\ProgramData\Blackmagic Design\DaVinci Resolve\Support\Fusion\Fuses
- **Linux:** ~/.local/share/DaVinciResolve/Fusion/Fuses